Course Registration System

Overview

The Course Allotment System is designed to facilitate the management of student course selections and allotments. This system provides functionalities for student and admin signup, login, course management, and viewing of student details and allotments.

File Structure

- `main.c`: Entry point of the application. Manages the main menu and integrates different modules.
- `admin.c`: Contains functions related to admin operations such as signup, login, and password reset.
- `student.c`: Contains functions related to student operations such as signup, login, and viewing details.
- `utility.c`: Provides utility functions for handling common tasks like string manipulation, file operations, encryption, and validation.
- `course.c`: Handles course-related operations such as reading course details, adding/removing courses, and allotting courses to students.
- `common.h`: Contains common definitions and constants used across the project.
- `admin.h`: Header file for `admin.c`.
- `student.h`: Header file for `student.c`.
- `utility.h`: Header file for `utility.c`.
- `course.h`: Header file for `course.c`.

Modules and Functions

`main.c`

Manages the main menu and directs the flow to various functionalities like student and admin signup, login, and course selection.

Functions:

- `main()`: Entry point of the application.
- `mainMenu()`: Displays the main menu and handles user choices.
- `signupMenu()`: Displays the signup menu for admin and student registration.
- `loginMenu()`: Displays the login menu for admin and student login.
- `resetPassword()`: Displays the menu for resetting passwords for admin and student.

`student.c`

Handles student operations including signup, login, and viewing details.

Functions:

- `studentSignup(const char *filename)`: Handles student signup process, including username and password input, validation, and storage.
- `studentLogin(const char *filename)`: Manages student login process by validating credentials against stored data.
- `viewStudentDetails(const char *filename, const char *reg_number)`: Allows viewing details of a student based on their registration number.

`admin.c`

Handles admin operations including signup, login, and password reset.

Functions:

- `adminSignup(const char *filename)`: Handles admin signup process.
- `adminLogin()`: Manages admin login process.
- `adminMenu()`: Provides a menu for admin to manage courses, view student details, start/end course selection, and run the allotter.
- `adminResetPassword(const char *filename)`: Allows admin to reset their password.

`utility.c`

Provides utility functions for handling common tasks.

Functions:

- `validatePassword(const char *password)`: Validates the complexity of the password.
- `encryptPassword(char *password)`: Encrypts the password for secure storage.
- `usernameExists(const char *filename, const char *username)`: Checks if a username already exists in the given file.
- `authenticateUser(const char *filename, const char *username, const char *password)`: Authenticates the user by comparing the username and encrypted password with stored data.

`course.c`

Handles course-related operations including adding/removing courses, and allotting courses to students.

Functions:

- `addCourse()`: Allows admin to add a new course.
- `removeCourse()`: Allows admin to remove an existing course.
- `viewStudentDetails()`: Allows admin to view student details based on department and semester.
- `startCourseSelection()`: Starts the course selection process.
- `endCourseSelection()`: Ends the course selection process.
- `runAllotter()`: Runs the allotter to allocate courses to students based on their preferences and availability.

How to Run

1. Compile the program using a C compiler:

   ```
   bash
   Copy code
   gcc main.c admin.c student.c utility.c course.c -o course_allotment
   ```

2. Run the executable:

   ```
   bash
   Copy code
   ./course_allotment
   ```

3. Follow the on-screen menu to navigate through the functionalities.

Note

Make sure that the necessary CSV files (`admin.csv`, `student.csv`, `courses.csv`, etc.) are present in the same directory as the executable. These files will be used for storing and retrieving data.